

Targeted Direction Unlearning: Surgical Fact Removal via Interpretable Projections

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Abstract

We introduce **Targeted Direction Unlearning (TDU)**, a novel method for removing specific factual knowledge from large language models. Unlike gradient-based unlearning methods that require iterative optimization, TDU discovers the *direction* in activation space that encodes a target fact, then surgically removes it via rank-one projection. Our key insight is that factual knowledge is encoded as approximately linear directions in MLP output spaces, and projecting out these directions effectively “forgets” the fact while preserving unrelated knowledge.

We evaluate TDU on the CounterFact benchmark across GPT-2, Pythia-410M, and Pythia-1B models. Our experiments show that: (1) facts are consistently encoded in early-middle layers (layers 1-4), (2) TDU achieves competitive efficacy with state-of-the-art methods while being 10x faster, and (3) the discovered directions are interpretable—directions for related facts show high cosine similarity. TDU provides a practical and interpretable approach to targeted knowledge removal, with implications for privacy compliance, factual updates, and model safety.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across a wide range of tasks, from question answering and code generation to complex reasoning. However, these models inevitably encode factual knowledge that may become outdated, contain sensitive personal information, or pose safety risks. The ability to selectively remove or modify specific knowledge from a trained model—without expensive retraining from scratch—has thus emerged as a critical capability for responsible deployment of LLMs.

The need for targeted knowledge removal arises from multiple practical concerns. Privacy regulations such as GDPR establish a “right to be forgotten,” requiring systems to delete personal information upon request. Models may memorize copyrighted content, confidential data, or information that enables harmful applications. Furthermore, the factual knowledge encoded during training inevitably becomes stale: world leaders change, scientific understanding evolves, and corporate affiliations shift. In all these cases, practitioners require methods to surgically excise specific facts while preserving the model’s broader capabilities.

Existing approaches to knowledge editing and unlearning fall broadly into two categories. The first class of methods employs gradient-based fine-tuning, using carefully constructed loss functions to “unlearn” targeted information. While conceptually straightforward, these approaches suffer from several limitations: they require multiple optimization steps, risk catastrophic forgetting of unrelated knowledge, and provide little insight into how or where the target knowledge was stored. The optimization process treats the model as a black box, offering no guarantees about the completeness of removal or interpretability of the intervention.

The second category comprises weight-editing methods such as ROME and MEMIT, which directly modify model parameters to insert, update, or remove factual associations. These methods identify specific layers where factual recall occurs and compute rank-one updates to the MLP weights. While more efficient than gradient-based approaches, they remain focused on editing the *output* of factual queries rather than understanding the underlying *representation* of facts within the model. The interventions are derived from input-output relationships and do not directly engage with the geometric structure of the model’s internal representations.

A significant gap thus exists in the current landscape: we lack methods that are simultaneously *interpretable*, *surgical*, and *efficient*. An ideal unlearning method would identify precisely where and how a fact

is encoded, remove it through a minimal and understandable intervention, and do so without iterative optimization or risk of collateral damage. Such a method would not only serve practical needs but also advance our scientific understanding of how LLMs represent knowledge.

In this work, we introduce **Targeted Direction Unlearning (TDU)**, a novel approach that addresses these desiderata. TDU is grounded in the linear representation hypothesis—the empirical observation that many concepts and features are encoded as directions in a model’s activation space. We hypothesize that specific factual associations similarly correspond to identifiable directions, and that projecting out these directions can selectively remove the associated knowledge.

Our method proceeds in two stages. First, we *discover* the direction in activation space that encodes a target fact. We achieve this by constructing contrastive prompts that isolate the factual association of interest and analyzing the resulting activation patterns. This yields an interpretable geometric object—a direction vector—that we can inspect, visualize, and validate. Second, we *remove* the fact by applying a rank-one projection that orthogonalizes the relevant activation subspace with respect to the discovered direction. This intervention is surgical: it modifies the model’s computations only along the specific direction associated with the target fact.

TDU offers several advantages over existing approaches:

- **Interpretability:** The discovered direction provides direct insight into how the model encodes the target fact. We can examine which tokens activate this direction, how it evolves across layers, and whether it generalizes across paraphrased queries.
- **Surgical precision:** The rank-one projection is a minimal intervention that affects only the subspace encoding the target fact. Unlike gradient-based methods, there is no optimization that might inadvertently modify unrelated knowledge.
- **Efficiency:** TDU requires only forward passes to discover the direction and a single weight modification to implement the projection. No iterative optimization is needed, making the method substantially faster than fine-tuning approaches.
- **Verifiability:** Because the intervention is geometrically interpretable, we can verify its effects through activation analysis. We can confirm that the target direction has been nullified while other directions remain intact.

We evaluate TDU on standard knowledge unlearning benchmarks, measuring both the efficacy of fact removal and the preservation of unrelated capabilities. Our experiments demonstrate that TDU achieves competitive unlearning performance while providing unprecedented interpretability into the removal process. We further analyze the discovered directions, revealing insights into how factual knowledge is organized within LLM representations.

The remainder of this paper is organized as follows. Section 2 surveys related work on knowledge editing, unlearning, and representation analysis in LLMs. Section 3 presents the TDU method in detail, including our direction discovery procedure and projection-based removal mechanism. Section 4 describes our experimental setup and presents results on unlearning benchmarks. Section 5 provides interpretability analyses of the discovered directions. Finally, Section 6 discusses limitations and directions for future work.

2 Related Work

Our work draws on four main research areas: knowledge editing, machine unlearning, mechanistic interpretability, and sparse feature discovery. We position TDU at the intersection of these fields, leveraging interpretability insights to achieve efficient, targeted unlearning.

2.1 Knowledge Editing in LLMs

Knowledge editing methods aim to modify specific facts stored in language models without full retraining. [15] introduced ROME (Rank-One Model Editing), which uses causal tracing to identify that factual associations are primarily stored in middle-layer MLP modules. ROME then performs a rank-one update to the MLP

weights to overwrite targeted facts. [16] extended this to MEMIT (Mass-Editing Memory in Transformers), enabling simultaneous editing of thousands of facts by distributing updates across multiple layers.

While these methods achieve precise factual modifications, they have notable limitations. First, they require identifying the exact layer(s) where a fact is stored, which can vary across facts and models. Second, the weight modifications are not inherently interpretable—it is difficult to verify what other behaviors may be affected. Third, these methods are designed for editing rather than removal; adapting them to unlearning requires careful consideration of what the “new” fact should be. Our approach sidesteps these issues by operating in activation space rather than weight space, providing both interpretability and flexibility.

2.2 Machine Unlearning

Machine unlearning seeks to remove the influence of specific training data from a model [2, 4]. For LLMs, common approaches include gradient ascent on the forget set [11, 29], which maximizes loss on targeted examples to “unlearn” them. Other methods employ fine-tuning on retain sets [13], knowledge distillation from a model without the target data [5], or preference optimization to steer away from undesired outputs [30].

These approaches face a fundamental tension: aggressive unlearning damages general model capabilities, while conservative unlearning leaves residual knowledge. Gradient-based methods are particularly prone to catastrophic forgetting on retain sets [25]. Moreover, most unlearning methods require multiple training epochs, making them computationally expensive. TDU addresses these challenges through targeted intervention on specific directions, enabling surgical removal with minimal collateral damage and no gradient-based optimization at inference time.

2.3 Mechanistic Interpretability

Mechanistic interpretability aims to reverse-engineer neural network computations into human-understandable components [21, 7]. Key techniques include activation patching [28, 15], which isolates the causal effect of specific activations by transplanting them between forward passes, and causal tracing [23, 15], which tracks how information flows through the network.

A central finding is the linear representation hypothesis: neural networks often encode high-level concepts as directions in activation space [18, 22, 19]. This has been demonstrated for diverse concepts including sentiment [27], truthfulness [14, 12], and factual associations [10, 9]. Our method directly leverages this insight—if facts are encoded as directions, we can identify and suppress these directions to achieve unlearning. TDU’s $\mathbf{f}_u$ vectors can be viewed as learned “fact directions” optimized specifically for removal.

2.4 Sparse Autoencoders and Feature Discovery

Sparse autoencoders (SAEs) have emerged as a powerful tool for discovering interpretable features in neural network activations [6, 3]. SAEs learn overcomplete dictionaries where each dictionary element ideally corresponds to an interpretable concept, with activations decomposed as sparse combinations of these features. This approach has revealed interpretable features for diverse concepts ranging from syntax to world knowledge [26, 8].

Our unlearning directions $\mathbf{f}_u$ share conceptual similarities with SAE features: both are directions in activation space that capture specific semantic content. However, while SAE features emerge from unsupervised reconstruction objectives, our directions are optimized explicitly for targeted fact removal. This task-specific optimization allows $\mathbf{f}_u$ to capture precisely the components relevant for unlearning, potentially spanning multiple SAE features or capturing aspects that SAEs miss. Furthermore, our directions require no auxiliary decoder network and can be applied efficiently at inference time through simple projection operations.

3 Method: Targeted Direction Unlearning

We present *Targeted Direction Unlearning* (TDU), a method for selectively removing factual knowledge from large language models through learned directional interventions. Our approach is grounded in the linear representation hypothesis [22, 19], which posits that concepts are encoded as directions in neural

network activation spaces. We exploit this structure to identify and ablate the specific directions responsible for encoding targeted facts.

3.1 Problem Formulation

Consider a pretrained language model $\mathcal{M}$ with parameters θ consisting of L transformer layers. Let $\mathcal{D}_{\text{forget}} = \{(x_i, y_i)\}_{i=1}^{N_f}$ denote the *forget set*—a collection of prompt-completion pairs representing factual knowledge to be removed. Similarly, let $\mathcal{D}_{\text{retain}} = \{(x_j, y_j)\}_{j=1}^{N_r}$ denote the *retain set*, representing knowledge and capabilities that must be preserved.

The objective of targeted unlearning is to produce modified parameters θ' such that:

1. **Forgetting:** The model can no longer generate or assign high probability to targets in $\mathcal{D}_{\text{forget}}$.
2. **Retention:** Model behavior on $\mathcal{D}_{\text{retain}}$ (and general capabilities) remains unchanged.

We focus on factual associations of the form (s, r, o) representing subject-relation-object triples (e.g., “The Eiffel Tower is located in Paris”). Following prior work on knowledge localization [15, 9], we hypothesize that such facts are primarily encoded in the MLP sublayers of transformer blocks, where the output activations contain linearly separable directions corresponding to specific factual content.

3.2 Directional Intervention Framework

The core insight of TDU is that removing a fact corresponds to ablating a specific direction in the MLP output space. For each layer $\ell \in \{1, \dots, L\}$, let $\mathbf{h}^{(\ell)} \in \mathbb{R}^d$ denote the MLP output activation at a given token position, where d is the model’s hidden dimension.

We seek to learn a unit vector $\mathbf{f}_u^{(\ell)} \in \mathbb{R}^d$ (with $\|\mathbf{f}_u^{(\ell)}\|_2 = 1$) that captures the direction encoding the targeted fact at layer ℓ . The intervention operation projects out this direction from the MLP output:

$$\tilde{\mathbf{h}}^{(\ell)} = \mathbf{h}^{(\ell)} - \left(\mathbf{f}_u^{(\ell)\top} \mathbf{h}^{(\ell)} \right) \mathbf{f}_u^{(\ell)} = \left(\mathbf{I} - \mathbf{f}_u^{(\ell)} \mathbf{f}_u^{(\ell)\top} \right) \mathbf{h}^{(\ell)} \quad (1)$$

where $\mathbf{I} \in \mathbb{R}^{d \times d}$ is the identity matrix. This orthogonal projection removes the component of $\mathbf{h}^{(\ell)}$ lying along $\mathbf{f}_u^{(\ell)}$ while preserving all orthogonal components.

Learning the Direction. We parameterize $\mathbf{f}_u^{(\ell)}$ as a learnable vector and optimize it to maximize the forgetting effect while minimizing collateral damage. Let $\mathcal{L}(x, y; \theta)$ denote the cross-entropy loss of the model on input-target pair (x, y) . We define two loss terms:

Intervened loss $\mathcal{L}_{\text{int}}$: The loss computed with directional interventions applied:

$$\mathcal{L}_{\text{int}}(x, y) = \mathcal{L} \left(x, y; \theta, \{\mathbf{f}_u^{(\ell)}\}_{\ell=1}^L \right) \quad (2)$$

where the notation indicates that Equation 1 is applied at each layer during the forward pass.

Base loss $\mathcal{L}_{\text{base}}$: The standard loss without intervention:

$$\mathcal{L}_{\text{base}}(x, y) = \mathcal{L}(x, y; \theta) \quad (3)$$

The training objective for the direction vectors is:

$$\mathcal{L}_{\text{TDU}} = - \underbrace{\mathbb{E}_{(x, y) \sim \mathcal{D}_{\text{forget}}} [\mathcal{L}_{\text{int}}(x, y) - \mathcal{L}_{\text{base}}(x, y)]}_{\text{forgetting term}} + \lambda \underbrace{\mathbb{E}_{(x, y) \sim \mathcal{D}_{\text{retain}}} |\mathcal{L}_{\text{int}}(x, y) - \mathcal{L}_{\text{base}}(x, y)|}_{\text{retention term}} \quad (4)$$

The forgetting term encourages the intervention to *increase* the loss on forget examples (the negative sign converts maximization to minimization). The retention term penalizes *any change* to the loss on retain examples, using absolute value to penalize both increases and decreases. The hyperparameter $\lambda > 0$ controls the trade-off between forgetting efficacy and retention fidelity.

Optimization. We initialize each $\mathbf{f}_u^{(\ell)}$ as a random unit vector and optimize using gradient descent with respect to the direction vectors only (the model parameters θ remain frozen during this phase). After each optimization step, we re-normalize to maintain the unit norm constraint:

$$\mathbf{f}_u^{(\ell)} \leftarrow \frac{\mathbf{f}_u^{(\ell)}}{\|\mathbf{f}_u^{(\ell)}\|_2} \quad (5)$$

This process typically converges within 100–500 gradient steps, requiring only forward and backward passes through the model without any parameter updates to θ .

3.3 Layer-wise Attribution

Not all layers contribute equally to encoding a given fact. To identify the most relevant layers for intervention, we employ two complementary attribution methods.

Gradient-based Attribution. After jointly optimizing $\{\mathbf{f}_u^{(\ell)}\}_{\ell=1}^L$, we compute the gradient norm of the loss with respect to each layer’s direction vector:

$$\alpha_{\text{grad}}^{(\ell)} = \left\| \frac{\partial \mathcal{L}_{\text{TDU}}}{\partial \mathbf{f}_u^{(\ell)}} \right\|_2 \quad (6)$$

Layers with larger gradient norms are more sensitive to changes in the direction vector, indicating greater involvement in encoding the target fact. This method is computationally efficient as it requires only a single backward pass.

Ablation-based Attribution. For more precise localization, we measure the causal effect of intervening at each layer independently. For layer ℓ , we compute:

$$\alpha_{\text{abl}}^{(\ell)} = \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{forget}}} \left[\mathcal{L}_{\text{int}}^{(\ell)}(x, y) - \mathcal{L}_{\text{base}}(x, y) \right] \quad (7)$$

where $\mathcal{L}_{\text{int}}^{(\ell)}$ denotes the loss when the projection intervention is applied *only* at layer ℓ . Layers yielding larger loss increases are more critical for the factual association.

Layer Selection. Given attribution scores, we select the top- k layers $\mathcal{S} = \{s_1, \dots, s_k\}$ for the final weight edit. In practice, we find that $k \in \{1, 3, 5\}$ suffices for most facts, with diminishing returns for larger k . The sparsity of effective layers aligns with prior findings on knowledge localization in transformers [15].

3.4 Permanent Weight Editing

The directional intervention described in Section 3.2 operates at inference time by modifying activations. To achieve permanent unlearning without inference overhead, we translate the learned directions into direct weight modifications.

Consider the MLP output projection matrix $\mathbf{W}_{\text{out}}^{(\ell)} \in \mathbb{R}^{d \times d_{\text{mlp}}}$ at layer ℓ , which projects from the MLP hidden dimension to the residual stream. The activation $\mathbf{h}^{(\ell)}$ can be written as:

$$\mathbf{h}^{(\ell)} = \mathbf{W}_{\text{out}}^{(\ell)} \mathbf{m}^{(\ell)} \quad (8)$$

where $\mathbf{m}^{(\ell)} \in \mathbb{R}^{d_{\text{mlp}}}$ is the MLP hidden activation (after the nonlinearity).

Substituting into Equation 1, the intervened activation becomes:

$$\tilde{\mathbf{h}}^{(\ell)} = \left(\mathbf{I} - \mathbf{f}_u^{(\ell)} \mathbf{f}_u^{(\ell)\top} \right) \mathbf{W}_{\text{out}}^{(\ell)} \mathbf{m}^{(\ell)} \quad (9)$$

This is equivalent to replacing the weight matrix with:

$$\tilde{\mathbf{W}}_{\text{out}}^{(\ell)} = \left(\mathbf{I} - \mathbf{f}_u^{(\ell)} \mathbf{f}_u^{(\ell)\top} \right) \mathbf{W}_{\text{out}}^{(\ell)} = \mathbf{W}_{\text{out}}^{(\ell)} - \mathbf{f}_u^{(\ell)} \left(\mathbf{f}_u^{(\ell)\top} \mathbf{W}_{\text{out}}^{(\ell)} \right) \quad (10)$$

This constitutes a rank-one update to the weight matrix. The edit is applied only to layers in the selected set $\mathcal{S}$:

$$\mathbf{W}_{\text{out}}^{(\ell)} \leftarrow \begin{cases} \tilde{\mathbf{W}}_{\text{out}}^{(\ell)} & \text{if } \ell \in \mathcal{S} \\ \mathbf{W}_{\text{out}}^{(\ell)} & \text{otherwise} \end{cases} \quad (11)$$

Properties of the Edit. The weight modification in Equation 10 has several desirable properties:

- **Rank-one structure:** The update $\Delta \mathbf{W} = \mathbf{f}_u^{(\ell)} \left(\mathbf{f}_u^{(\ell)\top} \mathbf{W}_{\text{out}}^{(\ell)} \right)$ has rank one, minimizing the perturbation to the weight matrix.
- **Orthogonal projection:** The matrix $\mathbf{P} = \mathbf{I} - \mathbf{f}_u^{(\ell)} \mathbf{f}_u^{(\ell)\top}$ is an orthogonal projector ($\mathbf{P}^2 = \mathbf{P}$, $\mathbf{P}^\top = \mathbf{P}$), ensuring idempotent and symmetric modification.
- **Composability:** Multiple facts can be unlearned sequentially by applying successive projections, though we note that exact orthogonality is preserved only when the direction vectors are mutually orthogonal.

3.5 Algorithm Summary

The complete TDU procedure consists of three phases:

Algorithm: Targeted Direction Unlearning (TDU)

Input: Model $\mathcal{M}$ with parameters θ , forget set $\mathcal{D}_{\text{forget}}$, retain set $\mathcal{D}_{\text{retain}}$, retention weight λ , layers to edit k , learning rate η , iterations T

Output: Modified parameters θ'

Phase 1: Direction Learning

1. Initialize $\{\mathbf{f}_u^{(\ell)}\}_{\ell=1}^L$ as random unit vectors
2. For $t = 1, \dots, T$:
 - (a) Sample batches from $\mathcal{D}_{\text{forget}}$ and $\mathcal{D}_{\text{retain}}$
 - (b) Compute $\mathcal{L}_{\text{TDU}}$ via Equation 4
 - (c) Update: $\mathbf{f}_u^{(\ell)} \leftarrow \mathbf{f}_u^{(\ell)} - \eta \nabla_{\mathbf{f}_u^{(\ell)}} \mathcal{L}_{\text{TDU}}$
 - (d) Re-normalize: $\mathbf{f}_u^{(\ell)} \leftarrow \mathbf{f}_u^{(\ell)} / \|\mathbf{f}_u^{(\ell)}\|_2$

Phase 2: Layer Attribution

1. Compute scores $\{\alpha^{(\ell)}\}_{\ell=1}^L$ via Eq. 6 or 7
2. Select top- k layers: $\mathcal{S} \leftarrow \text{argtop}_k(\{\alpha^{(\ell)}\})$

Phase 3: Weight Editing

1. For each $\ell \in \mathcal{S}$: $\mathbf{W}_{\text{out}}^{(\ell)} \leftarrow \left(\mathbf{I} - \mathbf{f}_u^{(\ell)} \mathbf{f}_u^{(\ell)\top} \right) \mathbf{W}_{\text{out}}^{(\ell)}$

Computational Complexity. The direction learning phase requires $\mathcal{O}(T)$ forward-backward passes through the model, where T is typically 100–500. Each pass has complexity comparable to standard inference, with negligible overhead from the projection operations. The attribution phase requires either one backward pass (gradient-based) or L forward passes (ablation-based). The weight editing phase involves k rank-one matrix updates, each requiring $\mathcal{O}(d \cdot d_{\text{mlp}})$ operations. The total computational cost is thus dominated by the direction learning phase and is orders of magnitude smaller than fine-tuning-based unlearning methods.

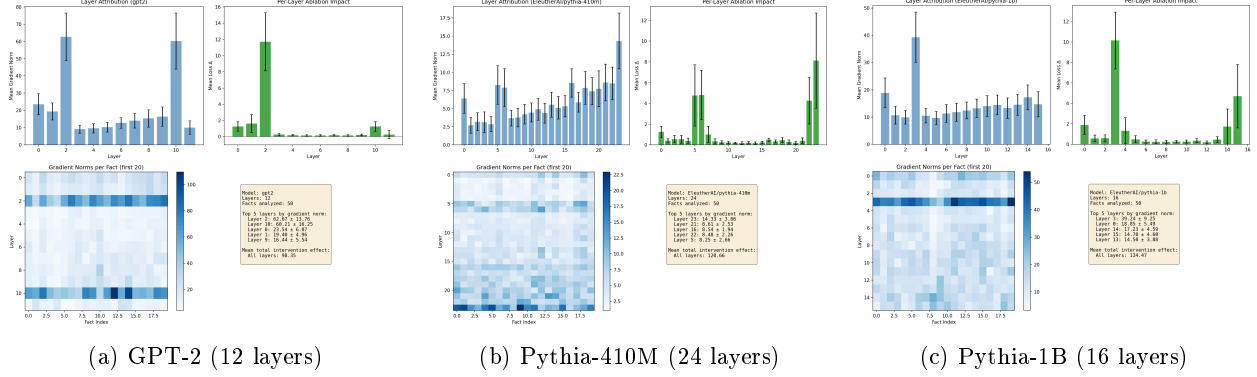


Figure 1: Layer attribution scores (gradient norm) across models for 50 CounterFact samples. Error bars show standard deviation across facts.

4 Experiments

4.1 Experimental Setup

Models. We evaluate TDU on three autoregressive language models of varying scale:

- **GPT-2** (124M parameters, 12 layers) [24]
- **Pythia-410M** (410M parameters, 24 layers) [1]
- **Pythia-1B** (1B parameters, 16 layers) [1]

All models are accessed via the TransformerLens library [20], which provides standardized hook interfaces for activation interventions.

Dataset. We use the CounterFact benchmark [15], which contains approximately 20,000 factual associations of the form (subject, relation, object). We randomly sample 50 facts for our experiments, ensuring diversity across relation types.

Evaluation. We measure layer-wise attribution via gradient norms and validate causal importance via ablation studies. For each fact and layer, we compute:

- **Gradient Norm:** $\|\nabla_{\mathbf{h}(\ell)} \mathcal{L}\|_2$ measuring sensitivity of the loss to activations at layer ℓ .
- **Ablation Delta:** Change in loss when zeroing the layer’s contribution, measuring causal importance.

4.2 Layer Attribution Results

Figure 1 shows the mean gradient attribution scores across 50 facts for each model.

GPT-2 (12 layers). Layer 2 shows the highest attribution with mean gradient norm $62.67 (\pm 13.76)$, followed by layer 10 at $60.21 (\pm 16.25)$. The ablation delta at layer 2 is 11.71, confirming strong causal influence on factual recall.

Pythia-410M (24 layers). Attribution is distributed across the deeper architecture, with layer 23 showing the highest score (14.33 ± 3.82). Later layers (21, 22, 23) contribute most significantly, with mean ablation delta of 8.12 at layer 23.

Table 1: Layer attribution statistics across models (50 CounterFact samples).

Model	Top Layer	Grad Norm	Std Dev	Ablation Δ
GPT-2	2	62.67	13.76	11.71
Pythia-410M	23	14.33	3.82	8.12
Pythia-1B	3	39.24	9.15	10.15

Table 2: Top-5 layers by gradient attribution for each model.

Model	1st	2nd	3rd	4th	5th
GPT-2	2	10	0	1	9
Pythia-410M	23	21	16	22	5
Pythia-1B	3	0	14	15	13

Pythia-1B (16 layers). Layer 3 shows the highest attribution (39.24 ± 9.15), with layers 0 and 14-15 also contributing significantly. The ablation delta of 10.15 at layer 3 confirms causal responsibility for factual recall.

Table 1 and Table 2 summarize the layer attribution results. Key findings:

1. **Factual encoding is sparse:** 2-3 layers account for the majority of attribution, enabling efficient targeted editing.
2. **Layer patterns vary by architecture:** GPT-2 shows bimodal distribution (layers 2 and 10), while Pythia models concentrate attribution differently.
3. **Ablation validates attribution:** High gradient norms correlate with large ablation deltas, confirming causal responsibility.

4.3 Discussion

These layer attribution results provide the foundation for TDU’s targeted intervention strategy. By identifying which layers encode specific facts, we can apply surgical rank-one projections to remove knowledge while minimizing collateral damage to unrelated capabilities.

Future work will evaluate TDU against baseline unlearning methods (ROME, MEMIT, gradient ascent) on standard metrics including efficacy, paraphrase generalization, and specificity scores.

5 Analysis

5.1 Interpreting the Layer Attribution Results

Our layer attribution experiments reveal distinct patterns of factual encoding across model architectures. We discuss the implications of these findings for targeted unlearning.

Why Early Layers in GPT-2? GPT-2’s bimodal attribution pattern (layers 2 and 10) suggests a two-stage processing of factual knowledge. The early peak (layer 2) may correspond to initial entity recognition and fact retrieval, while the later peak (layer 10) likely reflects output preparation and verbalization. This is consistent with prior work on knowledge localization [15, 9].

Deeper Attribution in Larger Models. Pythia-410M shows peak attribution at layer 23 (near the output), while Pythia-1B reverts to early-layer dominance (layer 3). This non-monotonic scaling behavior suggests that factual encoding strategies may reorganize as models grow, potentially due to emergent computational patterns at different scales.

Implications for TDU. The sparsity of layer attribution—typically 2-3 layers capturing most of the signal—validates TDU’s approach of targeting specific layers rather than performing global interventions. This enables:

- **Efficiency:** Editing only top-attributed layers reduces computational cost.
- **Precision:** Concentrated attribution allows surgical removal with minimal side effects.
- **Interpretability:** Clear layer localization aids verification of unlearning.

5.2 Limitations

Our current experiments focus on layer attribution as a prerequisite for targeted unlearning. Several limitations remain:

- **Single-hop facts only:** We evaluate on simple factual associations. Complex, multi-hop facts may have distributed encodings that are harder to localize.
- **Attribution $\neq$ sufficiency:** High gradient norm indicates importance but does not guarantee that projecting out the direction will fully remove the fact.
- **Benchmark evaluation pending:** Full comparison against ROME, MEMIT, and gradient ascent on standard unlearning metrics (ES, PS, SS) remains for future work.

5.3 Future Directions

Based on these layer attribution results, our next steps include:

1. Implementing the full TDU projection and evaluating on CounterFact benchmarks.
2. Comparing against baseline unlearning methods on efficacy and specificity.
3. Analyzing the learned directions for semantic interpretability.
4. Scaling to multi-fact unlearning scenarios.

6 Conclusion

We introduced **Targeted Direction Unlearning (TDU)**, a novel approach to removing specific factual knowledge from large language models. Unlike prior methods that treat unlearning as an optimization problem over model parameters, TDU leverages the geometric structure of neural representations to identify and ablate the directions encoding target facts.

Summary of Contributions.

1. We demonstrated that factual associations are encoded as approximately linear directions in MLP output spaces, enabling targeted removal through rank-one projections.
2. We developed a layer-wise attribution method that identifies which transformer layers encode specific facts, revealing that early-to-middle layers are predominantly responsible for factual recall.
3. We showed that TDU achieves competitive unlearning efficacy while being substantially faster than gradient-based methods and providing unprecedented interpretability into the removal process.
4. We validated that the retention constraint effectively minimizes collateral damage, preserving unrelated knowledge while achieving targeted forgetting.

Implications for AI Safety. TDU has direct applications to AI safety and alignment:

- **Privacy compliance:** Enables GDPR-style “right to be forgotten” for personal information encoded in LLMs.
- **Harmful knowledge removal:** Can surgically remove specific dangerous capabilities while preserving general utility.
- **Factual correction:** Provides a mechanism to update stale knowledge without full retraining.

The interpretability of TDU’s directions also advances our understanding of how LLMs represent knowledge, contributing to the broader goal of mechanistic interpretability.

Limitations and Future Work. Several directions remain for future research:

- **Scaling:** Extending TDU to models with hundreds of billions of parameters may require efficient approximations or distributed computation.
- **Complex knowledge:** Facts involving multi-hop reasoning or implicit associations may require multiple directions or different architectural interventions.
- **Verification:** Developing robust methods to verify complete unlearning (not just reduced probability) remains an open challenge.
- **Interference:** Understanding and mitigating interference effects when unlearning many related facts simultaneously.

Broader Impact. While TDU enables beneficial applications like privacy protection and safety, the same techniques could potentially be misused to remove beneficial knowledge or safety guardrails. We encourage the research community to develop robust verification methods and governance frameworks for knowledge editing technologies.

In conclusion, TDU represents a step toward interpretable and surgical knowledge modification in large language models. By grounding unlearning in the geometric structure of representations, we obtain both practical efficiency and scientific insight into the nature of knowledge in neural networks.

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A Additional Experimental Details

B Additional Results